

```
In [5]: pip install sweetviz
```

Collecting sweetviz

Downloading sweetviz-2.3.3-py3-none-any.whl.metadata (22 kB)

Requirement already satisfied: pandas!=1.0.0,!=1.0.1,!=1.0.2,>=0.25.3 in c:\users\kaish\anaconda3\envs\comp_9721_may\lib\site-packages (from sweetviz) (2.3.0)

Requirement already satisfied: numpy>=1.16.0 in c:\users\kaish\anaconda3\envs\comp_9721_may\lib\site-packages (from sweetviz) (2.3.0)

Requirement already satisfied: matplotlib>=3.1.3 in c:\users\kaish\anaconda3\envs\comp_9721_may\lib\site-packages (from sweetviz) (3.10.0)

Requirement already satisfied: tqdm>=4.43.0 in c:\users\kaish\anaconda3\envs\comp_9721_may\lib\site-packages (from sweetviz) (4.67.3)

Requirement already satisfied: scipy>=1.3.2 in c:\users\kaish\anaconda3\envs\comp_9721_may\lib\site-packages (from sweetviz) (1.15.3)

Requirement already satisfied: jinja2>=2.11.1 in c:\users\kaish\anaconda3\envs\comp_9721_may\lib\site-packages (from sweetviz) (3.1.6)

Collecting importlib_resources>=1.2.0 (from sweetviz)

Downloading importlib_resources-7.1.0-py3-none-any.whl.metadata (4.0 kB)

Requirement already satisfied: MarkupSafe>=2.0 in c:\users\kaish\anaconda3\envs\comp_9721_may\lib\site-packages (from jinja2>=2.11.1->sweetviz) (3.0.2)

Requirement already satisfied: contourpy>=1.0.1 in c:\users\kaish\anaconda3\envs\comp_9721_may\lib\site-packages (from matplotlib>=3.1.3->sweetviz) (1.3.2)

Requirement already satisfied: cyclor>=0.10 in c:\users\kaish\anaconda3\envs\comp_9721_may\lib\site-packages (from matplotlib>=3.1.3->sweetviz) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in c:\users\kaish\anaconda3\envs\comp_9721_may\lib\site-packages (from matplotlib>=3.1.3->sweetviz) (4.58.4)

Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\kaish\anaconda3\envs\comp_9721_may\lib\site-packages (from matplotlib>=3.1.3->sweetviz) (1.4.8)

Requirement already satisfied: packaging>=20.0 in c:\users\kaish\anaconda3\envs\comp_9721_may\lib\site-packages (from matplotlib>=3.1.3->sweetviz) (25.0)

Requirement already satisfied: pillow>=8 in c:\users\kaish\anaconda3\envs\comp_9721_may\lib\site-packages (from matplotlib>=3.1.3->sweetviz) (11.2.1)

Requirement already satisfied: pyparsing>=2.3.1 in c:\users\kaish\anaconda3\envs\comp_9721_may\lib\site-packages (from matplotlib>=3.1.3->sweetviz) (3.2.3)

Requirement already satisfied: python-dateutil>=2.7 in c:\users\kaish\anaconda3\envs\comp_9721_may\lib\site-packages (from matplotlib>=3.1.3->sweetviz) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in c:\users\kaish\anaconda3\envs\comp_9721_may\lib\site-packages (from pandas!=1.0.0,!=1.0.1,!=1.0.2,>=0.25.3->sweetviz) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in c:\users\kaish\anaconda3\envs\comp_9721_may\lib\site-packages (from pandas!=1.0.0,!=1.0.1,!=1.0.2,>=0.25.3->sweetviz) (2025.2)

Requirement already satisfied: six>=1.5 in c:\users\kaish\anaconda3\envs\comp_9721_may\lib\site-packages (from python-dateutil>=2.7->matplotlib>=3.1.3->sweetviz) (1.17.0)

Requirement already satisfied: colorama in c:\users\kaish\anaconda3\envs\comp_9721_may\lib\site-packages (from tqdm>=4.43.0->sweetviz) (0.4.6)

Downloading sweetviz-2.3.3-py3-none-any.whl (15.1 MB)

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Downloading importlib_resources-7.1.0-py3-none-any.whl (37 kB)

Installing collected packages: importlib_resources, sweetviz

```

----- 0/2 [importlib_resources]
----- 1/2 [sweetviz]
----- 1/2 [sweetviz]
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----- 2/2 [sweetviz]

```

Successfully installed importlib_resources-7.1.0 sweetviz-2.3.3

Note: you may need to restart the kernel to use updated packages.

```
In [11]: import pandas as pd

df = pd.read_csv(r"C:\Users\kaish\OneDrive\Desktop\My Tmu\CIND820 Big Data Analytic
df.head(50)
```

Out[11]:

	male	age	education	currentSmoker	cigsPerDay	BPMeds	prevalentStroke	prevalent
0	1	39	4.0	0	0.0	0.0	0	
1	0	46	2.0	0	0.0	0.0	0	
2	1	48	1.0	1	20.0	0.0	0	
3	0	61	3.0	1	30.0	0.0	0	
4	0	46	3.0	1	23.0	0.0	0	
5	0	43	2.0	0	0.0	0.0	0	
6	0	63	1.0	0	0.0	0.0	0	
7	0	45	2.0	1	20.0	0.0	0	
8	1	52	1.0	0	0.0	0.0	0	
9	1	43	1.0	1	30.0	0.0	0	
10	0	50	1.0	0	0.0	0.0	0	
11	0	43	2.0	0	0.0	0.0	0	
12	1	46	1.0	1	15.0	0.0	0	
13	0	41	3.0	0	0.0	1.0	0	
14	0	39	2.0	1	9.0	0.0	0	
15	0	38	2.0	1	20.0	0.0	0	
16	1	48	3.0	1	10.0	0.0	0	
17	0	46	2.0	1	20.0	0.0	0	
18	0	38	2.0	1	5.0	0.0	0	
19	1	41	2.0	0	0.0	0.0	0	
20	0	42	2.0	1	30.0	0.0	0	
21	0	43	1.0	0	0.0	0.0	0	
22	0	52	1.0	0	0.0	0.0	0	
23	0	52	3.0	1	20.0	0.0	0	
24	1	44	2.0	1	30.0	0.0	0	
25	1	47	4.0	1	20.0	0.0	0	
26	0	60	1.0	0	0.0	0.0	0	
27	1	35	2.0	1	20.0	0.0	0	
28	0	61	3.0	0	0.0	0.0	0	
29	0	60	1.0	0	0.0	0.0	0	

	male	age	education	currentSmoker	cigsPerDay	BPMeds	prevalentStroke	prevalent
30	1	36	4.0	1	35.0	0.0	0	
31	1	43	4.0	1	43.0	0.0	0	
32	0	59	1.0	0	0.0	0.0	0	
33	1	61	NaN	1	5.0	0.0	0	
34	1	54	1.0	1	20.0	0.0	0	
35	1	37	2.0	0	0.0	0.0	0	
36	1	56	NaN	0	0.0	0.0	0	
37	1	52	1.0	0	0.0	0.0	0	
38	0	42	1.0	1	1.0	0.0	0	
39	1	36	3.0	0	0.0	0.0	0	
40	0	43	2.0	1	10.0	0.0	0	
41	0	41	2.0	1	1.0	0.0	0	
42	0	52	1.0	0	0.0	1.0	0	
43	1	54	2.0	0	0.0	0.0	0	
44	0	53	3.0	0	0.0	1.0	0	
45	0	49	2.0	0	0.0	0.0	0	
46	0	65	1.0	0	0.0	0.0	0	
47	1	46	1.0	1	20.0	0.0	0	
48	0	63	2.0	1	40.0	0.0	0	
49	1	36	3.0	1	20.0	NaN	0	

```
In [13]: df.shape
```

```
Out[13]: (4240, 16)
```

4238 = number of patients (rows) 16 = features (columns)

```
In [15]: df.columns
```

```
Out[15]: Index(['male', 'age', 'education', 'currentSmoker', 'cigsPerDay', 'BPMeds',
               'prevalentStroke', 'prevalentHyp', 'diabetes', 'totChol', 'sysBP',
               'diaBP', 'BMI', 'heartRate', 'glucose', 'TenYearCHD'],
              dtype='object')
```

```
In [23]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4240 entries, 0 to 4239
Data columns (total 16 columns):
#   Column                Non-Null Count  Dtype
---  -
0   male                  4240 non-null   int64
1   age                   4240 non-null   int64
2   education             4135 non-null   float64
3   currentSmoker        4240 non-null   int64
4   cigsPerDay            4211 non-null   float64
5   BPMeds               4187 non-null   float64
6   prevalentStroke       4240 non-null   int64
7   prevalentHyp         4240 non-null   int64
8   diabetes              4240 non-null   int64
9   totChol              4190 non-null   float64
10  sysBP                4240 non-null   float64
11  diaBP                4240 non-null   float64
12  BMI                  4221 non-null   float64
13  heartRate            4239 non-null   float64
14  glucose              3852 non-null   float64
15  TenYearCHD           4240 non-null   int64
dtypes: float64(9), int64(7)
memory usage: 530.1 KB

```

```
In [25]: df.isnull().sum()
```

```

Out[25]: male                0
age                0
education         105
currentSmoker     0
cigsPerDay        29
BPMeds            53
prevalentStroke   0
prevalentHyp      0
diabetes          0
totChol           50
sysBP             0
diaBP             0
BMI               19
heartRate         1
glucose           388
TenYearCHD        0
dtype: int64

```

```
In [31]: df['TenYearCHD'].value_counts()
```

```

Out[31]: TenYearCHD
0      3596
1       644
Name: count, dtype: int64

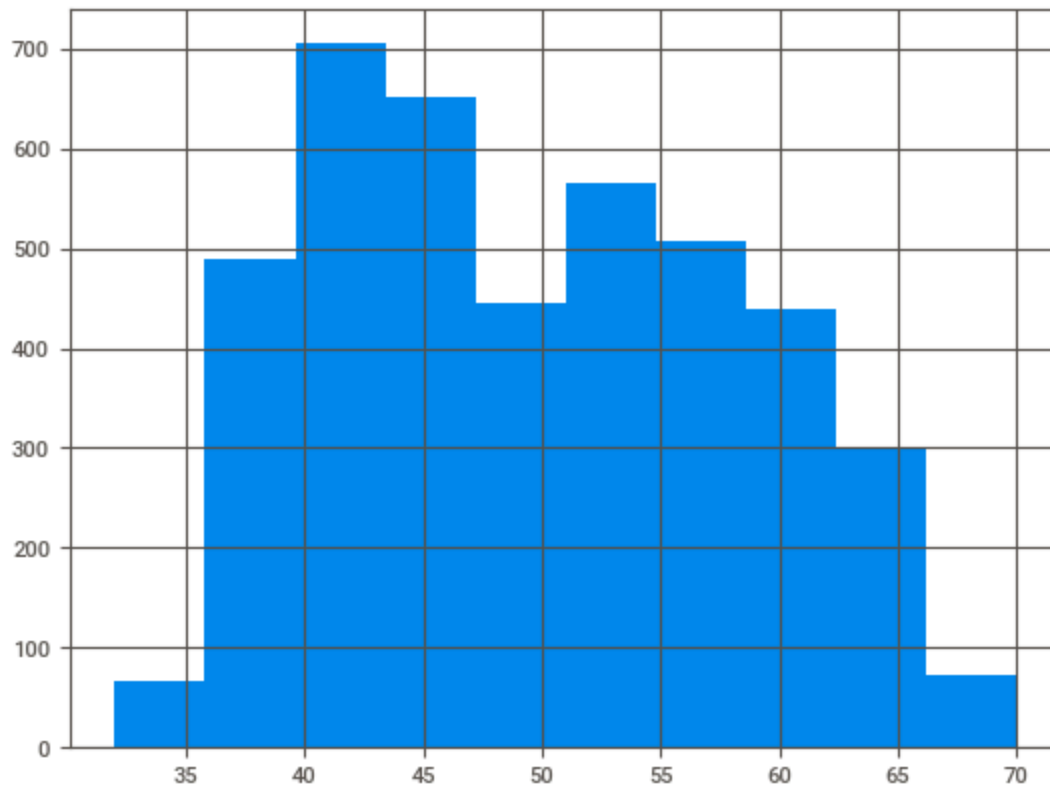
```

```
In [33]: df['TenYearCHD'].value_counts(normalize=True)
```

```
Out[33]: TenYearCHD
0      0.848113
1      0.151887
Name: proportion, dtype: float64
```

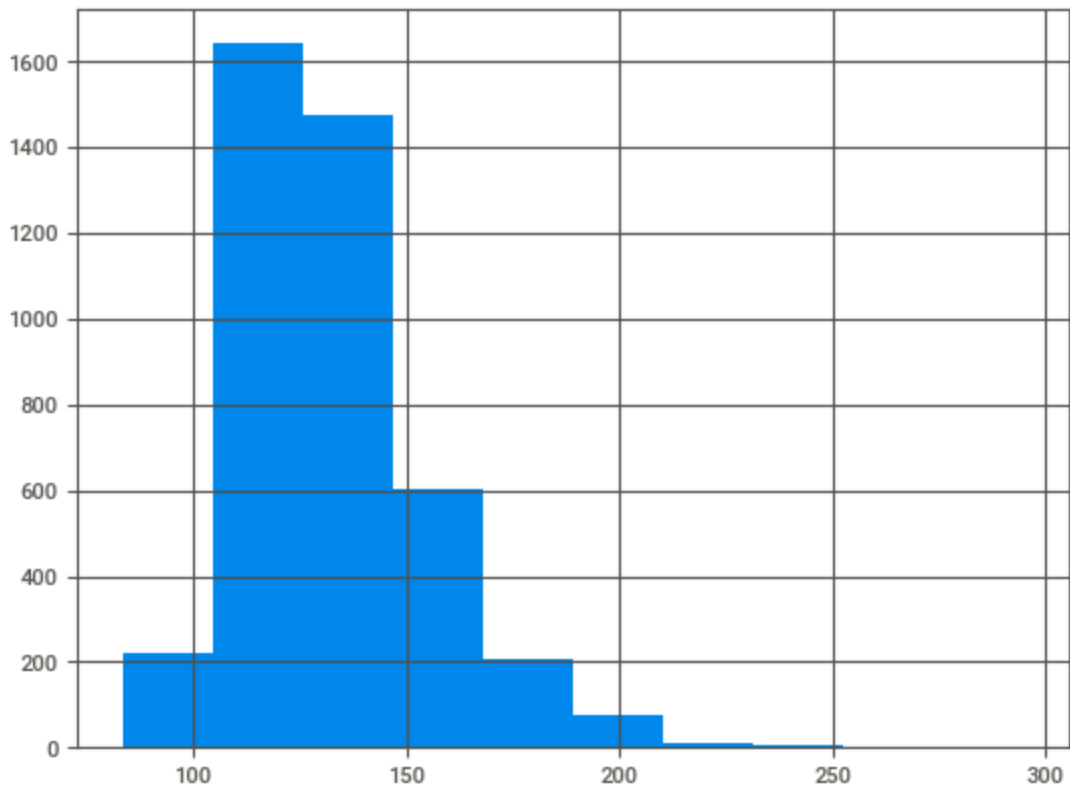
```
In [35]: df['age'].hist()
```

```
Out[35]: <Axes: >
```



```
In [37]: df['sysBP'].hist()
```

```
Out[37]: <Axes: >
```

```
In [39]: df.skew(numeric_only=True)
```

```
Out[39]: male          0.285997
age            0.228867
education      0.688730
currentSmoker  0.023595
cigsPerDay     1.247052
BPMeds         5.551462
prevalentStroke 12.912160
prevalentHyp   0.818827
diabetes       5.995907
totChol        0.871881
sysBP         1.145285
diaBP         0.713250
BMI            0.982183
heartRate     0.644372
glucose       6.214948
TenYearCHD    1.940516
dtype: float64
```

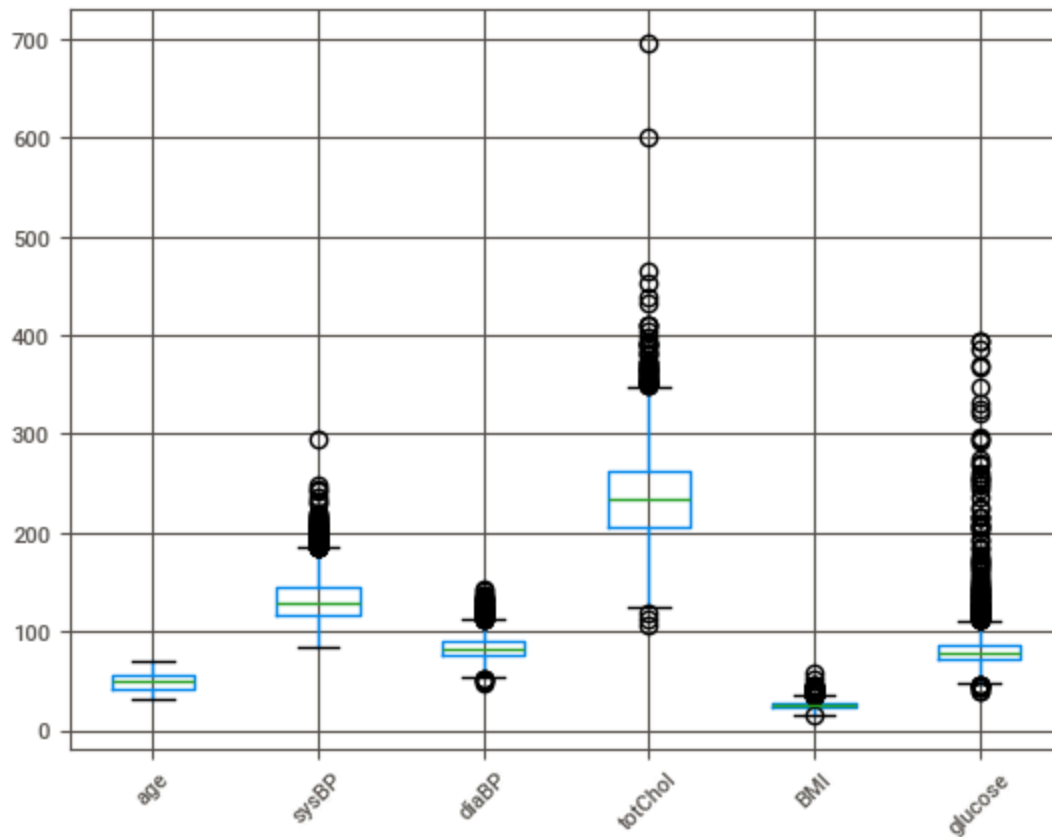
```
In [41]: df.corr()
```

Out[41]:

	male	age	education	currentSmoker	cigsPerDay	BPMeds	pr
male	1.000000	-0.029014	0.017415	0.197026	0.317143	-0.052504	
age	-0.029014	1.000000	-0.166356	-0.213662	-0.192959	0.123052	
education	0.017415	-0.166356	1.000000	0.018528	0.008323	-0.010898	
currentSmoker	0.197026	-0.213662	0.018528	1.000000	0.769774	-0.048927	
cigsPerDay	0.317143	-0.192959	0.008323	0.769774	1.000000	-0.046155	
BPMeds	-0.052504	0.123052	-0.010898	-0.048927	-0.046155	1.000000	
prevalentStroke	-0.004550	0.057679	-0.035142	-0.032980	-0.032711	0.117370	
prevalentHyp	0.005853	0.306799	-0.081726	-0.103710	-0.066645	0.261067	
diabetes	0.015693	0.101314	-0.038749	-0.044285	-0.037089	0.052060	
totChol	-0.070413	0.262554	-0.023613	-0.046488	-0.026479	0.080623	
sysBP	-0.035879	0.394053	-0.129496	-0.130281	-0.088797	0.254194	
diaBP	0.058199	0.205586	-0.061919	-0.107933	-0.056715	0.194122	
BMI	0.081871	0.136096	-0.137747	-0.167857	-0.093293	0.100702	
heartRate	-0.116932	-0.012843	-0.054182	0.062686	0.075564	0.015230	
glucose	0.005979	0.122356	-0.035843	-0.056726	-0.058886	0.051197	
TenYearCHD	0.088374	0.225408	-0.054248	0.019448	0.057755	0.087519	

In [43]: `import matplotlib.pyplot as plt`

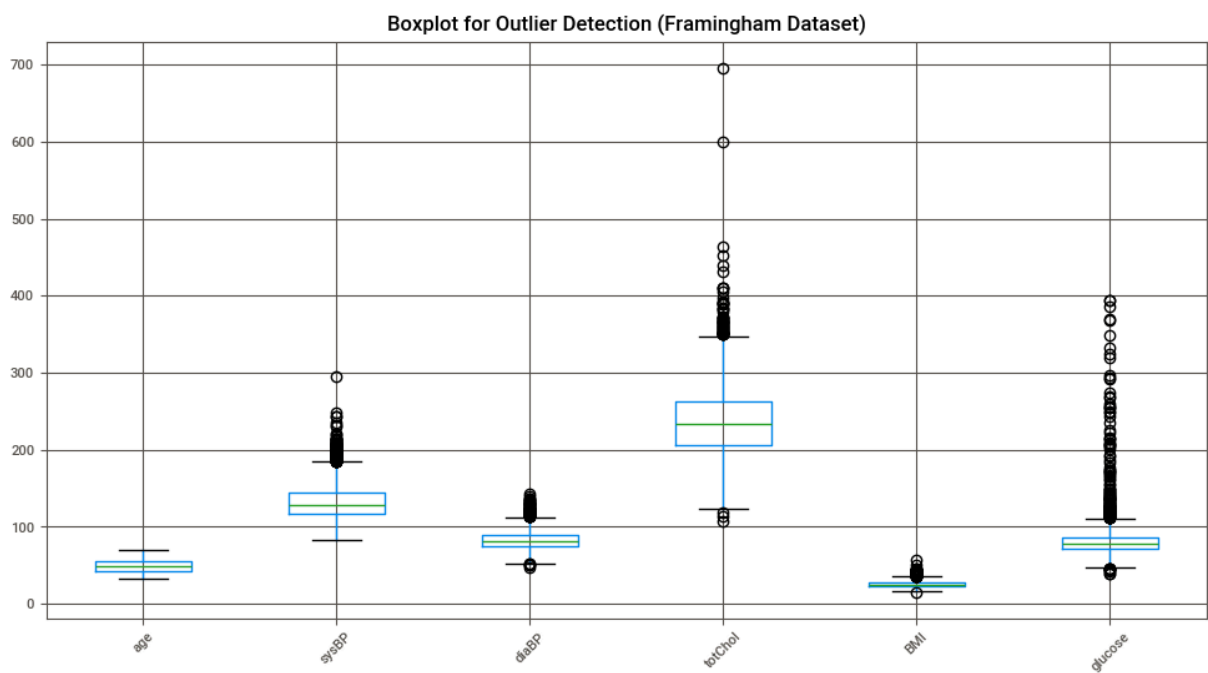
```
df[['age', 'sysBP', 'diaBP', 'totChol', 'BMI', 'glucose']].boxplot()
plt.xticks(rotation=45)
plt.show()
```



```
In [45]: import matplotlib.pyplot as plt

cols = ['age', 'sysBP', 'diaBP', 'totChol', 'BMI', 'glucose']

df[cols].boxplot(figsize=(12,6))
plt.xticks(rotation=45)
plt.title("Boxplot for Outlier Detection (Framingham Dataset)")
plt.show()
```



In [47]: `import sweetviz as sv`

```
report = sv.analyze(df)
report.show_html("framingham_report.html")
```

| [0%] 00:00 ->...

Report framingham_report.html was generated! NOTEBOOK/COLAB USERS: the web browser MAY not pop up, regardless, the report IS saved in your notebook/colab files.

The exploratory data analysis reveals that the Framingham dataset is suitable for predictive modeling but requires careful preprocessing, including handling missing values, addressing class imbalance, managing skewed distributions, and reducing redundancy among correlated features. These preprocessing steps will be essential before building machine learning models for heart disease prediction.

In []: